

ELECTRICITY AND CIRCUIT LAWS

Electricity can be made to travel in a flow, or current. Electric current can be generated by a source of electromotive force (EMF), such as a battery cell, and directed around wire loops called circuits to power electrical devices. There are

two main ways in which flow is driven around a circuit: an alternating current (AC), which flows to and fro; and direct current, which flows in one direction only. Particular laws govern the way that a current flows around a circuit.

| LAW                     | DESCRIPTION                                                                                                                                                                        | FORMULA                                           | KEY                                                                                                                           |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Coulomb's law           | The force of attraction or repulsion between two charged particles is directly proportional to the product of the charges and inversely proportional to the distance between them. | $F = k \frac{q_1 q_2}{r^2}$                       | k = Coulomb's constant<br>q <sub>1</sub> and q <sub>2</sub> = electric (point) charges<br>r <sup>2</sup> = square of distance |
| Ohm's law               | This law expresses the relationships between voltage, resistance, and current, which can be expressed in several ways.                                                             | $I = \frac{V}{R} \quad R = \frac{V}{I}$ $V = I R$ | R = resistance<br>I = current<br>V = potential difference (voltage)                                                           |
| Kirchhoff's current law | The sum of the electric currents entering any junction in a circuit is equal to the sum of those leaving the junction.                                                             | $\sum I = 0$                                      | $\sum$ = summation symbol<br>I = current                                                                                      |
| Kirchhoff's voltage law | The sum of the voltage changes around the path of any closed loop is zero.                                                                                                         | $\sum V = 0$                                      | $\sum$ = summation symbol<br>V = potential difference (voltage)                                                               |

SUBATOMIC PARTICLES

These are the basic building blocks of matter. Physicists distinguish between elementary particles (with no substructure) and composite particles (composed of smaller structures). Elementary particles are the fundamental constituents of everything in the Universe. Scientists think that every particle is twinned with an opposing antiparticle, which if paired can annihilate each other, producing packages of light (photons).

| ELEMENTARY PARTICLES                                                                                                                                                                                | COMPOSITE PARTICLES (HADRONS)                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Quarks</b> Particles that make protons and neutrons. There are six "flavors": up, down, charm, strange, top, and bottom.                                                                         | <b>Baryons</b> Particles made up of three quarks. The best known are the proton (two up and one down quarks) and the neutron (one up and two down quarks)                                              |
| <b>Leptons</b> A group of six particles, comprising electron, muon, and tau particles and their associated neutrinos (the electron-neutrino, muon-neutrino, and tau-neutrino).                      |                                                                                                                                                                                                        |
| <b>Gauge bosons</b> Particles associated with the four fundamental forces (see below). No gauge boson has yet been found for gravity, although the existence of a "graviton" has been hypothesized. | <b>Mesons</b> Particles made from a quark with an antiquark. There are many types, including the positive pion (up quark with down antiquark) and the negative kaon (strange quark with up antiquark). |

THE FOUR FUNDAMENTAL FORCES

All matter in the Universe is subject to four basic forces: gravity, electromagnetism, and the strong and weak nuclear forces. Each one is associated with a subatomic "messenger" particle. Physicists theorize that the forces were once unified in a single force that split in the first fraction of a second after the Big Bang. Matter particles affected by a particular force produce and absorb specific force carriers.

| PARTICLE    | FORCE                | RELATIVE STRENGTH | RANGE IN (M)      |
|-------------|----------------------|-------------------|-------------------|
| Graviton    | Gravity              | 10 <sup>-41</sup> | Infinite          |
| Photon      | Electromagnetic      | 1                 | Infinite force    |
| Gluon       | Strong nuclear force | 25                | 10 <sup>-15</sup> |
| W, Z bosons | Weak nuclear force   | 0.8               | 10 <sup>-18</sup> |

COMMON EQUATIONS

The following equations are commonly used in physics.

| QUANTITY                       | STATEMENT                                                   | FORMULA                           |
|--------------------------------|-------------------------------------------------------------|-----------------------------------|
| Kinetic energy                 | ½ mass × velocity <sup>2</sup>                              | E <sub>k</sub> = ½mv <sup>2</sup> |
| Weight                         | mass × gravitational field strength                         | W = mg                            |
| Power                          | work done or energy transferred<br>time taken               | P = $\frac{W}{t}$                 |
| Speed                          | distance moved<br>time taken                                | s = $\frac{d}{t}$                 |
| Velocity                       | displacement<br>time taken                                  | v = $\frac{s}{t}$                 |
| Acceleration                   | change in velocity<br>time taken for this change            | a = $\frac{(v-u)}{t}$             |
| Resultant force                | mass × acceleration                                         | F = ma                            |
| Momentum                       | mass × velocity                                             | m v                               |
| Refractive index               | speed of light in a vacuum<br>speed of light in a medium    | n = $\frac{c}{v}$                 |
| Wave speed                     | frequency × wavelength                                      | v = f λ                           |
| Electric charge                | current × time taken                                        | q = I t                           |
| Potential difference (voltage) | current × resistance or energy transferred<br>charge        | V = I R<br>V = $\frac{W}{q}$      |
| Resistance                     | voltage<br>current                                          | R = $\frac{V}{I}$                 |
| Electrical energy              | potential difference (or voltage)<br>× current × time taken | E = V I t                         |
| Work done                      | force × distance moved in the direction<br>of the force     | W = F s                           |
| Efficiency                     | work output<br>work input × 100%                            | $\frac{W_o}{W_i} \times 100\%$    |